

Fyn AI Phase 2 — Patch Notes

Version: v0.9.5 **Date:** 1 April 2026 **Branch:** `fynImprovement` (42 commits) **Stats:** 45 files changed, 7,097 insertions, 176 deletions **Tests:** 71 new tests (145 assertions), full regression 2,139/2,148 passing

System Prompt Refactor

The 670-line monolithic system prompt in `HasAiChat.php` has been decomposed into a **10-layer composable architecture** via `SystemPromptBuilder`:

Layer	Component	Type
1	Core Identity	STATIC — identity, security, scope, personality, response format
2	Compliance Rules	STATIC — FCA hedging language, 17 banned acronyms, no-icons, joint ownership
3	FCA Process	STATIC — 6-step advice process, tool usage, data creation guidance
4	User Profile	DYNAMIC — name, age, income, employment, family
5	Financial Position	DYNAMIC — net worth, module metrics, ranked recommendations
6	Existing Records	DYNAMIC — record summaries filtered by query type
7	Data Completeness	DYNAMIC — prerequisite gate status, navigation rules
7b	Review Due	DYNAMIC — data changes since last advice, annual review prompts
8	Query Knowledge	DYNAMIC — per-domain financial knowledge (RAG)
8b	Tools & Triggers	DYNAMIC — mandatory tool calls, decision tree triggers
9	KYC Status	DYNAMIC — data completeness check result (PASS/BLOCKED)
10	Page Context	DYNAMIC — current page context

Each layer is a separate class, independently testable, and assembled per-query based on classification.

Multi-Label Query Classification

Every user message is classified before the AI sees it. The `QueryClassifier` identifies a **primary type** plus **related types** — a pension question also triggers tax, savings, and affordability checks.

22 query types across 7 modules: protection (2), savings (3), retirement (3), investment (3), estate (2), goals (1), tax (1), plus property, income, holistic, general, `data_entry`, navigation, affordability.

Critical bypass: `data_entry` and `navigation` queries skip the entire FCA process — no KYC check, no knowledge injection, no mandatory tools. "I have a pension with £50,000" creates the record immediately.

"Take me to my property page" navigates instantly.

Implicit related types: Pension advice always adds tax + savings + affordability. Any "maximise" query adds emergency fund check. Holistic health queries check all modules.

KYC Gates — Data Check Before Advice

KycGateChecker verifies data completeness before Fyn gives advice:

- **Universal requirements:** date of birth, marital status, employment, income, expenditure
- **Module-specific:** protection needs dependants, investment needs risk profile, estate needs assets
- **Checks ALL classified modules** — pension + savings + affordability all checked

If data is missing, Fyn lists what's needed, offers to help enter it conversationally, and navigates to the **exact correct page** (not a guess — routes specified in the prompt).

Knowledge RAG

Instead of injecting all 1,800 tokens of financial knowledge into every prompt, **QueryKnowledge** returns only the domains relevant to the classified query:

- **Pension query:** pension + income + affordability knowledge (~1,328 tokens saved)
- **Data entry:** no knowledge at all (~3,109 tokens saved)
- **Holistic review:** all domains (unchanged)

Records and recommendations are also filtered by classification — a pension question only sees pension records and retirement recommendations.

Mandatory Tool Sequences

<required_tools> block injected per query type tells the AI which tools to call BEFORE responding:

- Pension questions → **get_tax_information(pension_allowances)**,
get_module_analysis(retirement)
 - Estate IHT questions → **get_tax_information(inheritance_tax)**,
get_module_analysis(estate)
 - Data entry → no mandatory tools
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Decision Tree Binding

<relevant_triggers> block maps each query type to the ActionDefinition triggers that should be referenced:

- Pension → **employer_match**, **contribution_increase**, **tax_relief**,
annual_allowance_exceeded
- Protection → **life_insurance_gap**, **income_protection_gap**, **critical_illness_gap**
- Holistic → all 51 triggers across all modules

Recommendations now include description, estimated saving (£ amounts), action steps, and trigger keys in the prompt — giving the AI specific calculation results to reference rather than inventing its own numbers.

Response Validation

StructuredResponseValidator checks every AI response before it reaches the user:

Check	Severity	Action
17 banned acronyms (IHT, CGT, SIPP, NRB, etc.)	High	Logged
Exposed record IDs (ID:123)	High	Stripped by sanitiser
Emoji / Unicode symbols	Medium	Logged
Planning jargon (waterfall, prioritise affordability)	Medium	Logged
Filler phrases (Certainly!, Of course!)	Low	Logged
Missing £ amounts in advice	High	Logged
HTML injection (script tags)	Critical	Stripped by sanitiser
Context block leaks ([Context: ...])	High	Stripped by sanitiser

Violations stored in message metadata for audit review.

Review System

- **ai_advice_logs table:** logs every advice interaction with query type, classification, KYC status, tools called, and user data snapshot (income, expenditure, employment, marital status at time of advice)
- **Data change detection:** **AdviceReviewService** compares current user data against the snapshot from their last advice — flags income changes >£1,000, expenditure changes >£100, employment/marital changes
- **Annual review prompts:** modules where advice is >12 months old get flagged in the system prompt — Fyn offers to review

AI Audit Dashboard

New admin tab at **/admin** → **AI Audit** — three-panel layout for full audit trail:

- **Left panel:** all users with AI conversations, searchable
- **Middle panel:** conversations for selected user, sorted by date
- **Right panel:** full message thread with:
 - User message and Fyn response
 - **Expandable system prompt** — the exact prompt sent to the LLM (stored per message)
 - **Tool calls** — which tools were called with inputs and results
 - **Classification** — query type + related types
 - **KYC status** — PASS/BLOCKED with missing items

- **Validation violations** — any compliance issues detected
 - **Token usage** — input/output token counts
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Admin Panel UI

Tab bar grouped into dropdowns:

- **Users** → User Metrics, User Management
- **AI** → AI Audit, AI Provider

Cleaner navigation with fewer top-level tabs.

Other Fixes

- **Chat scroll:** user message scrolls to top on send, rolling status messages while Fyn thinks
- **Suggestion bypass:** suggestions now send directly, bypassing canSend reactivity issue
- **Income field:** removed "Other Income" — all income must be categorised by type
- **Affordability rules:** added to Fyn's knowledge base — checks surplus, emergency fund, debt before recommending contributions
- **Response quality:** removed irrelevant concepts, enforced specific £ amounts, banned planning jargon
- **UserMetricsServiceTest:** fixed date edge case on month boundaries
- **xAI token tracking:** added `stream_options.include_usage` for token counts